RV College of Engineering ®

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DENSENET: Densely Connected Convolutional Networks

INTRODUCTION

- Densely Connected Convolutional Networks (DenseNet) is a feed-forward convolutional neural network (CNN) architecture that links each layer to every other layer.
- Densenet is an Image classification Model. DenseNet overcome **vanishing gradient problem** and provide us high accuracy compared to other Deep Convolutional neural Networks by simply connecting every layer directly with each other.
- The vanishing gradient problem occurs in deep neural networks when the gradients become extremely small as they are propagated back through the layers. his causes the earlier layers in the network to learn very slowly or not at all.
- It’s especially common with activation functions like sigmoid or tanh.

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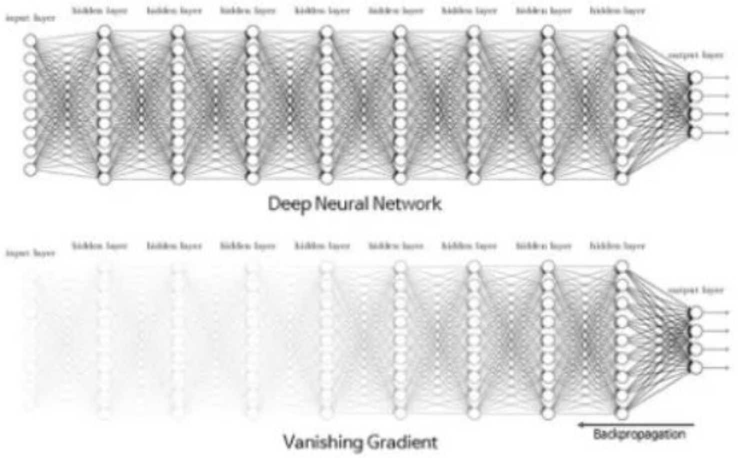
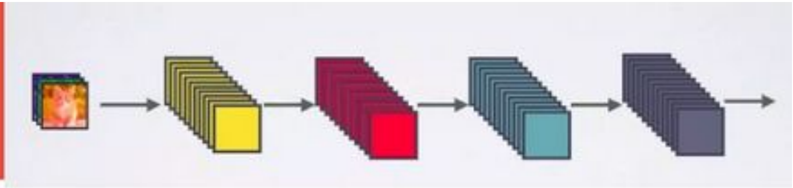
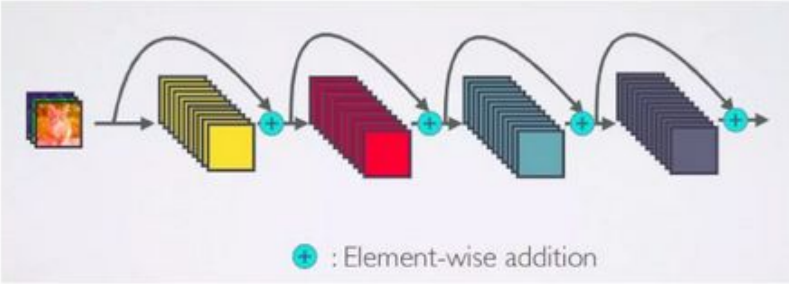
Vanishing Gradient Problem



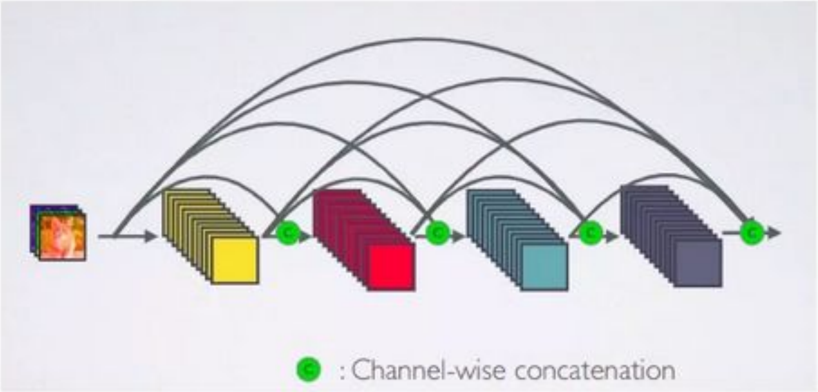
Traditional CNN



ResNet



DenseNet



DenseNet 121 Architecture

DenseNet-121 has the following layers:

- 1 7x7 Convolution
- 58 3x3 Convolution
- 61 1x1 Convolution
- 4 AvgPool
- 1 Fully Connected Layer

DenseNet 121 Architecture

1. Basic convolution layer with 64 filters of size 7X7 and a stride of 2
2. Basic pooling layer with 3x3 max pooling and a stride of 2
3. Dense Block 1 with 2 convolutions repeated 6 times
4. Transition layer 1 (1 Conv + 1 AvgPool)
5. Dense Block 2 with 2 convolutions repeated 12 times
6. Transition layer 2 (1 Conv + 1 AvgPool)
7. Dense Block 3 with 2 convolutions repeated 24 times
8. Transition layer 3 (1 Conv + 1 AvgPool)
9. Dense Block 4 with 2 convolutions repeated 16 times
10. Global Average Pooling layer- accepts all the feature maps of the network to perform classification
11. Output layer

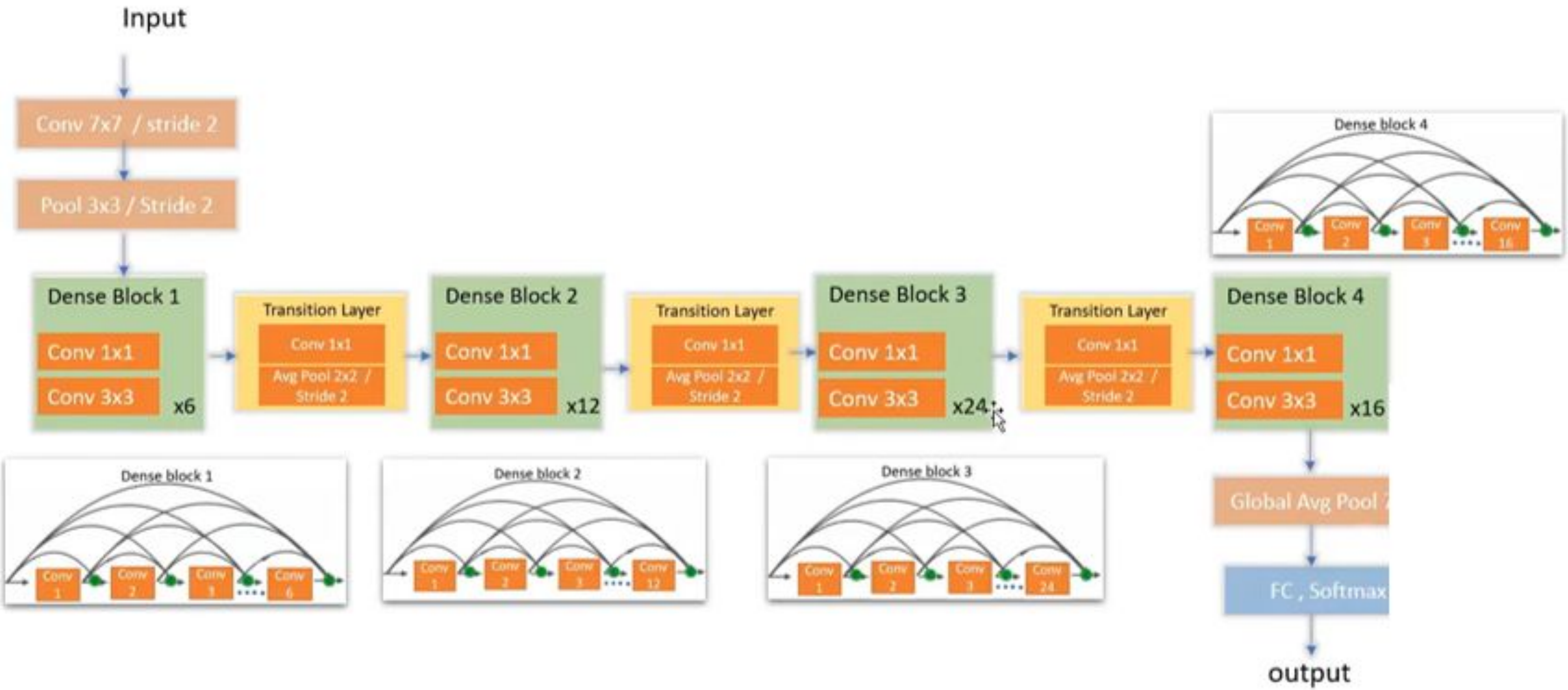
DenseNet 121 Architecture

- Input will go to a convolution neural network with a filter size of 7x7 using a stride of two. Then we have a pooling layer, and the filter size we are using over here is 3x3, and stride is two.
- In dense nets, we have dense blocks. You can see we have four green blocks called dense blocks. The layers between the blocks are called Transition Layers.
- In block 1,we have six convolution layers. Within a block Each layer is connected to every other layer
- Transition layers are used between dense blocks to control the size of the network in terms of both the number of feature maps (channels) and the spatial dimensions (height and width).
- Transition layers include a 1x1 convolution operation, which reduces the number of feature maps (channels).

DenseNet 121 Architecture

- Without transition layers, the concatenation of feature maps in each dense block would result in an exponential increase in the number of channels.
- The transition layer applies a 2x2 average pooling operation with a stride of 2, which halves the spatial dimensions (height and width) of the feature maps.
- This downs-ampling reduces the size of the feature maps progressively, making the network more computationally efficient and reducing memory usage.
- After the last dense block, a global average pooling layer reduces the spatial dimensions, providing a feature vector to the classifier. A fully connected layer follows, with softmax activation for classification.

DenseNet 121 Architecture

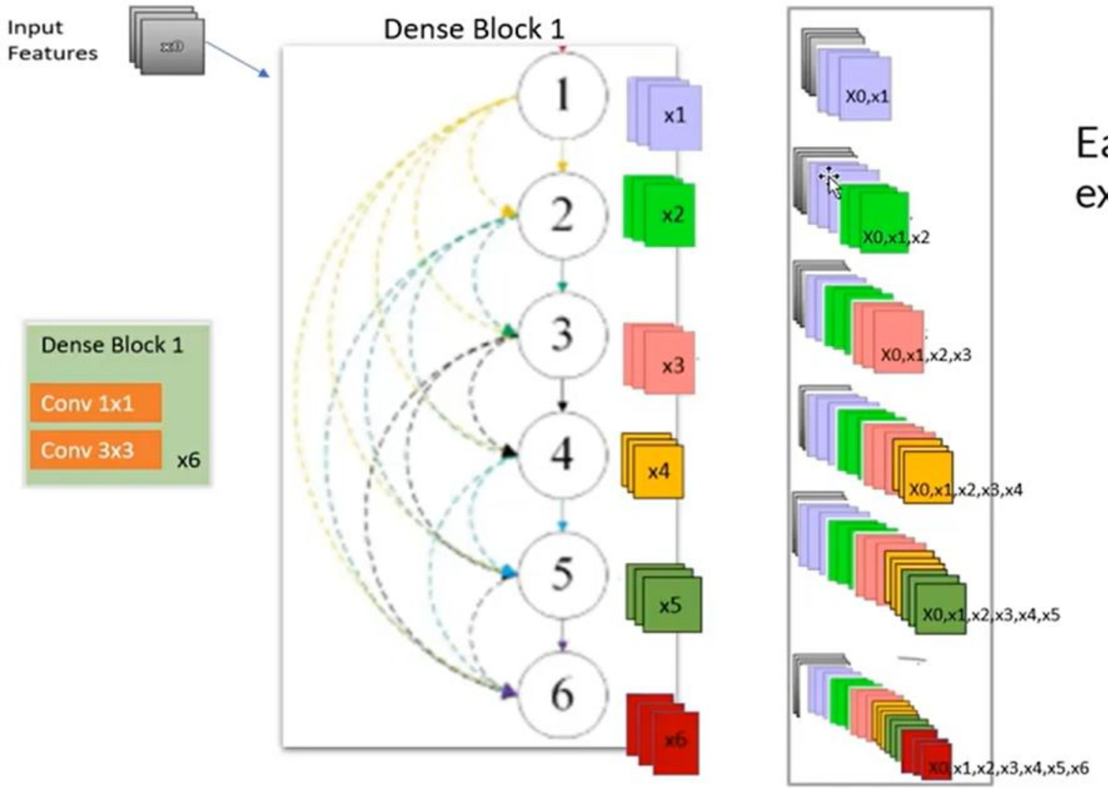


DenseNet 121 Architecture

- Components of DenseNet include:
  - DenseBlocks
  - Pooling layers
  - Growth Rate
  - Bottleneck layers

Inside Dense block

Ea



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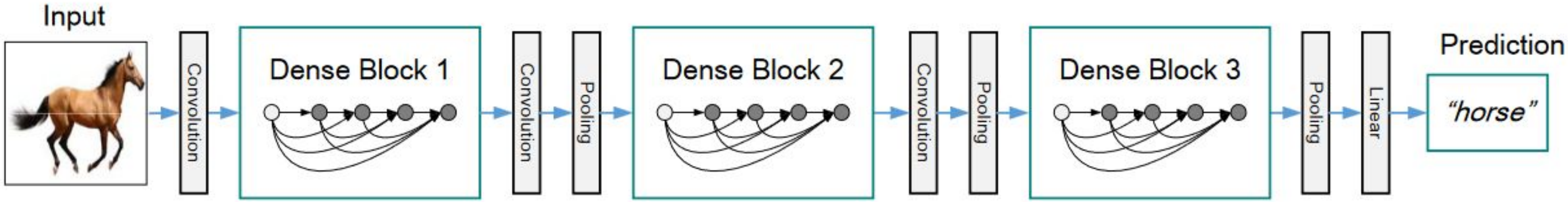
Dense Block

- A Dense Block is a key component in DenseNet (Dense Convolutional Network), where each layer is directly connected to every other layer within the same block. This means that the output of each layer is used as input for all subsequent layers within the block.
- The output of each layer is concatenated (not summed) with the input of all following layers within the block. Each layer in a dense block takes as input the feature maps of all preceding layers, and it passes its own feature maps to all subsequent layers. This results in a highly interconnected structure within the block.
- DenseNets concatenate the feature maps at each layer, ensuring that all feature maps are preserved throughout the network. The number of new feature maps produced by each layer is controlled by the growth rate. If the growth rate is set to 32, each layer will produce 32 new feature maps, which are then concatenated with all previous ones.

- Pooing Layer: The pooling layer can be either max-pool or average-pool and reduces the size of feature-maps. To facilitate down-sampling, the network is divided into multiple dense blocks with pooling layers inserted between them.



- Growth Rate: K is referred to as growth rate and regulates how much new information each layer contributes to global state.
- Bottleneck layer: reduce the number of input feature maps before applying the dense connections. They are typically 1x1 convolutions that reduce the computational cost

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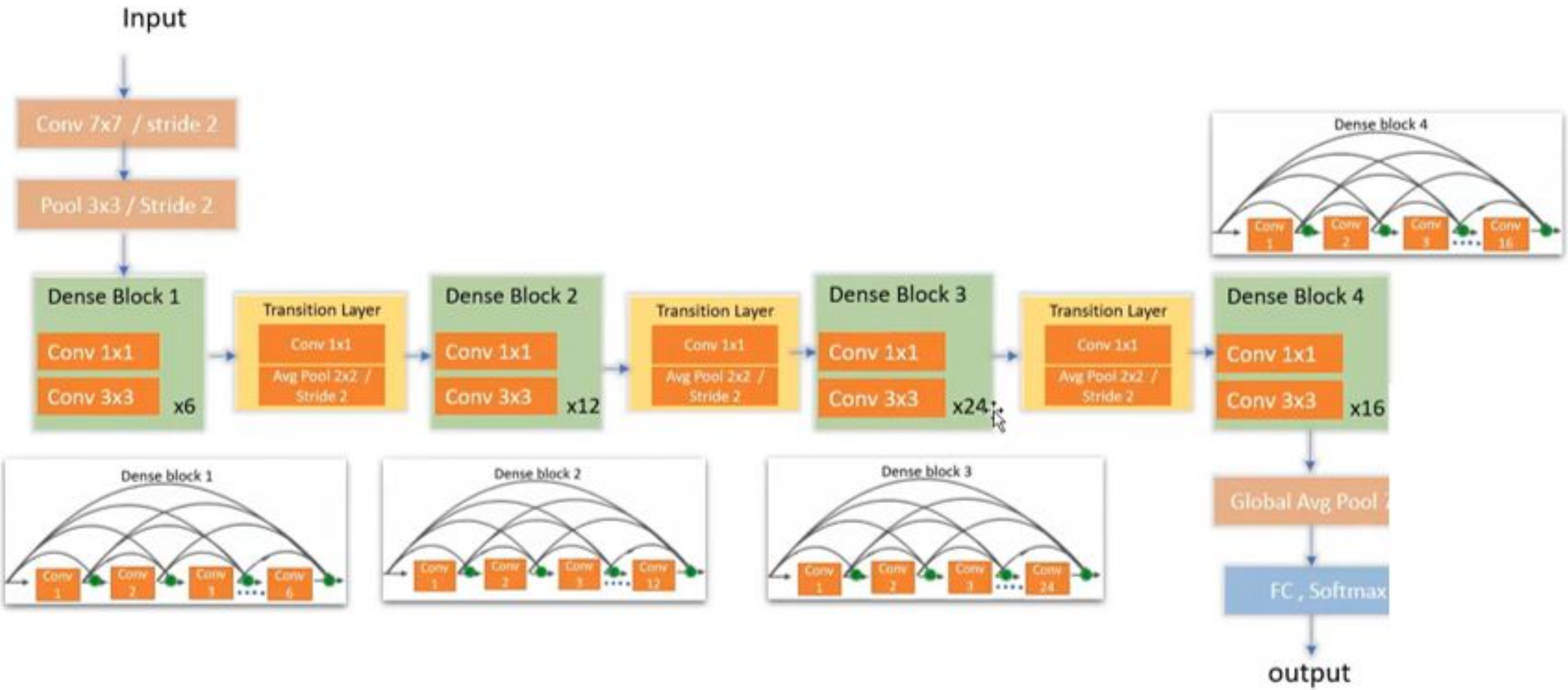
DenseNet 121 Architecture



Advantages

- Efficient Feature Reuse:
- Mitigating the Vanishing Gradient Problem
- Reduced Computational Load

Applications

- **NLP**: Used in translation, sentiment analysis, and text generation.
- **Generative Models**: Used as a generator in generative models, to produce new pictures.
- **Object Detection**: item recognition in photos and movies including automobiles, people, and buildings.
- **Medical Image**: To identify and categorize various types of cancers, lesions, and other anomalies.
- **Audio**: Implemented in audio processing applications including voice recognition, production, and audio synthesis.
- **Image**: Classifying photos into diverse categories such as wildlife, objects, and settings.
- **Semantic Segmentation**: Segment pictures into distinct areas such as sky, buildings, or roads.

THANKYOU